TUTORIAL-6

Ridge Regression
L2 Regularization

Property Features & Prices

Bedrooms (x_1)	Age (years) (x_2)	Price ($ 100000) (y)	$y-\hat{y}$
3	5	2	~~0.63~~
3	3	3	-0.37
			~~0.62~~
			-0.38
			0.1369
			0.1444

$$Y = MX + C$$

$$2 = C + 3m_1 + 5m_2$$

$$3 = c + 3m_1 + 3m_2$$

$$\underbrace{\begin{pmatrix} 2 \\ 3 \end{pmatrix}}_{Y} = \underbrace{\begin{pmatrix} 1 & 3 & 5 \\ 1 & 3 & 3 \end{pmatrix}}_{X} \underbrace{\begin{pmatrix} c \\ m_1 \\ m_2 \end{pmatrix}}_{w}$$

$$\lambda = 1$$

$$w = (X^T X + \lambda I)^{-1} (X^T y)$$

(Regularization)

$$\lambda = 1, \quad I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{3\times 3}$$

$$X^T X = \begin{pmatrix} 1 & 1 \\ 3 & 3 \\ 5 & 3 \end{pmatrix}_{3\times 2} \begin{pmatrix} 1 & 3 & 5 \\ 1 & 3 & 3 \end{pmatrix}_{2\times 3} = \begin{pmatrix} 1\cdot 1 + 1\cdot 1 & 1\cdot 3 + 1\cdot 3 & 1\cdot 5 + 1\cdot 3 \\ 3\cdot 1 + 3\cdot 1 & 3\cdot 3 + 3\cdot 3 & 3\cdot 5 + 3\cdot 3 \\ 5\cdot 1 + 3\cdot 1 & 5\cdot 3 + 3\cdot 3 & 5\cdot 5 + 3\cdot 3 \end{pmatrix}$$

$$\begin{pmatrix} 2 & 6 & 8 \\ 6 & 18 & 24 \\ 8 & 24 & 34 \end{pmatrix} + \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 3 & 6 & 8 \\ 6 & 19 & 24 \\ 8 & 24 & 35 \end{pmatrix}^{-1}$$

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$$X^T X = \begin{pmatrix} 3 & 6 & 8 \\ 6 & 19 & 24 \\ 8 & 24 & 35 \end{pmatrix}^{-1} = \frac{1}{95}\begin{pmatrix} 89 & -18 & -8 \\ -18 & 41 & -24 \\ -8 & -24 & 21 \end{pmatrix}$$

$$X^T y = \begin{pmatrix} 1 & 1 \\ 3 & 3 \\ 5 & 3 \end{pmatrix}\begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{matrix} 1.2+1.3 \\ \\ \\ \end{matrix} = \begin{pmatrix} 5 \\ 15 \\ 19 \end{pmatrix}$$

$$\begin{pmatrix} 23/95 \\ 69/95 \\ -1/95 \end{pmatrix} = \begin{pmatrix} 0.242 \\ 0.726 \\ -0.011 \end{pmatrix} \begin{matrix} \to c \\ \to m_1 \\ \to m_2 \end{matrix}$$

$$\hat{y}_{pred} = 0.2 + 0.7 x_1 - 0.01 x_2$$

$$\hat{y}_1 = 0.2 + 0.7(3) - 0.01(5)$$

$$= 2.37$$

$$\hat{y}_2 = 2.38$$

$$MSE = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2$$

$$= \frac{0.2813}{2}$$

$$= 0.25$$

$$R^2 = 1 - \frac{\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{N}(y_i - \bar{y})^2} = 0.042$$

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